PHP – Introduction

- Developed in 1994 by Apache Group.
- PHP: **H**ypertext **P**reprocessor.
- A server scripting language.
- Mainly used for form handling and database access.
- A powerful tool for making dynamic and interactive webpages.

PHP – Introduction

Advantages

- Very simple and easy to learn
- Interpreted Language
- Open source
- Platform Independent
- Can directly integrated with HTML
- Support to design dynamic web applications
- Reliable, efficient and flexible scripting language
- Provides support for file system, managing user sessions, cookies, E-mail Management, execute the system commands, create directories, etc.,

PHP – Introduction

Syntax

The code must be enclosed with in:

```
<?php
    // Statements (or) Codes
?>
```

- If anyone wants to refer file then ***"inlude"*** construct is used.

  Example: ***Include("myfile.html")***

Comments

```
// This is a single-line comment
# This is also a single-line comment
/*
This is a multiple-lines comment block
that spans over multiple lines
*/
```

Working Principle of PHP

- The most popular way of installing PHP is using XAMPP.
- XAMPP is a free distribution package that makes it easy to install Apache Web Server, MySQL, PHP, PEAR.

1. Go to the site: https://www.apachefriends.org/index.html
2. Click on download XAMPP for Windows or Linux depending on the OS
3. When prompted for the download, click "Save" and wait for the download finish.
4. Install the program and click on "Run". Accept default settings by clicking "Next" button. Finally, installation completion message is displayed.
5. On the drive, the XAMPP folder will be created. Click on xampp_start file, this will enable to start Apache, MySQL and Tomcat start.

Working Principle of PHP

6. The control panel for XAMPP will look like this,



6. Write a PHP script and save it in C:\XAMPP\htdocs\php-examples folder by giving the filename and extension as .php.
7. Open the web browser and type http://localhost/php-examples/filename.php.
8. The web application will be executed within the web browser.

PHP – Variables

- Variables are the entities that are used for storing the values.
- PHP is a dynamically typed language.
- **Syntax: *$variable_name = value;***
- If the value is not assigned to the variable then they by default the value is NULL.
- **Rules for Variables:**
  - The variable must start with letter or underscore _.
  - It consists of alphanumeric characters or underscore.
  - There should not be space in the name of the variable.
  - While assigning the values to the variable the variable must start with the $.
  - Example: ***$marks = 100;***

PHP – Data Types

- Four scalar data types used in PHP.
- **Integer:** To display the integer value. The size is 32 bits.
- **Boolean:** Only two types of values TRUE and FALSE.
- **Double:** To display the real values. It includes the number with decimal point, exponentiation or both.
- **String:** The string literals can be defined using either single or double quotes.

PHP – Constants

- Constant is an identifier that contains some value.
- Once the constant value is assigned to this identifier it ***does not get changed***.
- The constant identifiers are specified in ***upper case***.
- The valid constant name must ***start with letter or underscore***.
- Use ***define function*** to assign value to the constant.

Example:

```
<?php
        define("MYVALUE", "10");
        echo MYVALUE;
        define("1MYVALUE", "something");
        echo 1MYVALUE;
?>
```

PHP – print Vs. echo

- **print:** Function used to create simple unformatted output.

Example:

```
print “Hello <b> World!!! </b>”
print (100);
```

```
<?php
print "<h2>PHP !</h2>";
print "Hello World!<br>";
?>
```

PHP – print Vs. echo

- **echo:** Simple statement that can be used with or without parenthesis.

Example:

```
echo "Hello <b> World!!! </b>"
$a – 10;
echo $a;
```

print in PHP	echo in PHP
Print can only output one string	Echo can output one or more strings
Print is slower than echo	Echo is faster than print
Print always return 1	Echo does not return any value

PHP – Operators

Arithmetic Operators

Operator	Name	Example	Explanation
+	Addition	$a + $b	Sum of operands
-	Subtraction	$a - $b	Difference of operands
*	Multiplication	$a * $b	Product of operands
/	Division	$a / $b	Quotient of operands
%	Modulus	$a % $b	Remainder of operands
**	Exponentiation	$a ** $b	$a raised to the power $b

PHP – Operators

Relational Operators

Operator	Name	Example	Explanation
==	Equal	$a == $b	Return TRUE if $a is equal to $b
===	Identical	$a === $b	Return TRUE if $a is equal to $b, and they are of same data type
!==	Not identical	$a !== $b	Return TRUE if $a is not equal to $b, and they are not of same data type
!=	Not equal	$a != $b	Return TRUE if $a is not equal to $b
<>	Not equal	$a <> $b	Return TRUE if $a is not equal to $b
<	Less than	$a < $b	Return TRUE if $a is less than $b
<=	Less than or equal to	$a <= $b	Return TRUE if $a is less than or equal $b
>=	Greater than or equal to	$a >= $b	Return TRUE if $a is greater than or equal $b
<=>	Spaceship	$a <=>$b	Return -1 if $a is less than $b Return 0 if $a is equal $b Return 1 if $a is greater than $b

PHP – Operators

Bitwise Operators

Operator	Name	Example	Explanation
&	And	$a & $b	Bits that are 1 in both $a and $b are set to 1, otherwise 0.
|	Or (Inclusive or)	$a | $b	Bits that are 1 in either $a or $b are set to 1
^	Xor (Exclusive or)	$a ^ $b	Bits that are 1 in either $a or $b are set to 0.
~	Not	~$a	Bits that are 1 set to 0 and bits that are 0 are set to 1
<<	Shift left	$a << $b	Left shift the bits of operand $a $b steps
>>	Shift right	$a >> $b	Right shift the bits of $a operand by $b number of places

PHP – Operators

Incrementing/Decrementing Operators

Operator	Name	Example	Explanation
++	Increment	++$a	Increment the value of $a by one, then return $a
		$a++	Return $a, then increment the value of $a by one
--	decrement	--$a	Decrement the value of $a by one, then return $a
		$a--	Return $a, then decrement the value of $a by one

PHP – Operators

Logical Operators

Operator	Name	Example	Explanation
and	AND	$a and $b	Return TRUE if both $a and $b are true
or	OR	$a or $b	Return TRUE if either $a or $b is true
xor	XOR	$a xor $b	Return TRUE if either $ or $b is true but not both
!	NOT	! $a	Return TRUE if $a is not true
&&	AND	$a && $b	Return TRUE if either $a and $b are true
||	OR	$a || $b	Return TRUE if either $a or $b is true

PHP – Operators

Assignment Operators

Operator	Name	Example	Explanation
=	Assign	$a = $b	The value of right operand is assigned to the left operand.
+=	Add then Assign	$a += $b	Addition same as $a = $a + $b
-=	Subtract then Assign	$a -= $b	Subtraction same as $a = $a - $b
*=	Multiply then Assign	$a *= $b	Multiplication same as $a = $a * $b
/=	Divide then Assign (quotient)	$a /= $b	Find quotient same as $a = $a / $b
%=	Divide then Assign (remainder)	$a %= $b	Find remainder same as $a = $a % $b

PHP – Operators

String Operators

Operator	Name	Example	Explanation
.	Concatenation	$a . $b	Concatenate both $a and $b
.=	Concatenation and Assignment	$a .= $b	First concatenate $a and $b, then assign the concatenated string to $a, e.g. $a = $a . $b

PHP – Operators

Array Operators

Operator	Name	Example	Explanation
+	Union	$a + $y	Union of $a and $b
==	Equality	$a == $b	Return TRUE if $a and $b have same key/value pair
!=	Inequality	$a != $b	Return TRUE if $a is not equal to $b
===	Identity	$a === $b	Return TRUE if $a and $b have same key/value pair of same type in same order
!==	Non-Identity	$a !== $b	Return TRUE if $a is not identical to $b
<>	Inequality	$a <> $b	Return TRUE if $a is not equal to $b

PHP – Flow Control and Looping

Statement	Syntax
if-else	if (condition) statement; else statement;

Switch…Case	switch(expression) {
	  case 1:statements
	  break;
	  case 2: statements
	  break;
	  ….
	  default: statements
	}

break:	break;
continue:	continue;

PHP – Flow Control and Looping

Statement	Syntax

while	while(condition) {
	statements;
	}

do…while	do { Statements; } while(condition);

for	for(initialization; condition; increment){
	Statements;
	}

foreach	Used to iterate through all the elements of array.

PHP – Arrays

- Array is a collection of similar type of elements, but in PHP you can have the elements of mixed type together in single array.
- PHP array is an ordered map (contains value on the basis of key).

PHP Array Types

1. Indexed Array
2. Associative Array
3. Multidimensional Array

PHP – Arrays

Indexed Array:

PHP index is represented by number which starts from 0.

Use to store number, string and object in the PHP array.

All PHP array elements are assigned to an index number by default.

Example

```
<?php
        $season=array("summer","winter","spring","autumn");
        echo "Season are: $season[0], $season[1], $season[2] and $season[3]";
?>
```

PHP – Arrays

Indexed Array:

Example

```
<?php
        $season[0]="summer";
        $season[1]="winter";
        $season[2]="spring";
        $season[3]="autumn";
        echo "Season are: $season[0], $season[1], $season[2] and $season[3]";
?>
```

PHP – Arrays

Associative Array

- Associative arrays are the arrays with named keys.
- It is a kind of array with name and value pair.
- Can associate name with each array elements in PHP using => symbol

Example:

```
$salary=array("Sonoo"=>"350000","John"=>"450000","Kartik"=>"200000");
```

(OR)

```
$salary["Sonoo"]="350000";
$salary["John"]="450000";
$salary["Kartik"]="200000";
```

PHP – Arrays

Multidimensional Array

- PHP multidimensional array is also known as array of arrays.
- It allows to store tabular data in an array.
- PHP multidimensional array can be represented in the form of matrix which is represented by row * column.

Example

```
<?php
$emp = array
 (
 array(1,"sonoo",400000),
 array(2,"john",500000),
 array(3,"rahul",300000)
 );
for ($row = 0; $row < 3; $row++) {
 for ($col = 0; $col < 3; $col++) {
  echo $emp[$row][$col]." ";
 }
 echo "<br/>";
}
?>
```

PHP – Arrays

Function	Purpose
extract($arr)	The array keys becomes the variable name and the array values become the variable values
implode($arr)	Converts array into string
explode(delimiter, str1,limit)	Splits a string
array_flip($arr)	Used to exchange the keys with their associated values in array
current($arr), next($arr)	Traversing the array
sort(($arr)	Sort arrays in ascending order
rsort($arr)	Sort arrays in descending order
asort($arr)	Sort associative arrays in ascending order, according to the value
ksort($arr)	Sort associative arrays in ascending order, according to the key
array_change_key_case($arr, CASE_CASENAME)	Changes the case of all key of an array
array_chunk(($arr, value)	Splits array into chunks
array_reverse($arr)	Returns an array containing elements in reversed order
count($arr)	Counts all elements in an array
array_search("target",$arr)	Searches the specified value in an array. It returns key if search is successful.

PHP – Strings

- PHP string is a sequence of characters i.e., used to store and manipulate text.

Example

```
<?php
$str1="Hello World";
$str2="Using double \"quote\" with backslash inside double quoted string";
$str3="Using escape sequences \n in double quoted string";
echo "$str1 <br/> $str2 <br/> $str3";
?>
```

PHP – Strings

Function	Purpose
strlen(str1)	Used to return the length of a string.
strcmp(str1, str2)	Compares the two strings
strtolower(str1)	Converts the string to lower case
strtoupper(str1)	Converts the string to upper case
trim(str1)	Remove whitespace or other characters from the beginning and end of the string.
str_split(str1)	Used to split a string into an array
strrev(str1)	Used to reverse the string
substr(str1, start, length)	Return the part of a string
str_word_count(str1)	Returns the number of words in the string
strpos(str1, str2)	Searches for a specific text within a string
str_replace(str1, str2)	Replaces some characters with some other characters in a string

PHP – Functions

- PHP function is a piece of code that can be reused many times.
- It can take input as argument list and return value.

Syntax:

```
function functionname(){
//code to be executed
}
```

PHP – File Handling

- PHP File System allows us to create file, read file line by line, read file character by character, write file, append file, delete file and close file.
- **Syntax:**

```
<?php
        $handle = fopen("c:\\folder\\file.txt", "r");  //open file in read mode
        $contents = fread($handle, filesize($filename));  //read file
        $fp = fopen('data.txt', 'w');  //open file in write mode
        fwrite($fp, 'hello ');
        fwrite($fp, 'php file');
        unlink('data.txt');  //delete file
        fclose($handle);
?>
```

PHP – File Handling

- **PHP Open File:** fopen() function is used to open file or URL and returns resource.

Mode	Description
r	Opens file in **read-only** mode. It places the file pointer at the beginning of the file.
r+	Opens file in **read-write** mode. It places the file pointer at the beginning of the file.
w	Opens file in **write-only** mode. It places the file pointer to the beginning of the file and truncates the file to zero length. If file is not found, it creates a new file.
w+	Opens file in **read-write** mode. It places the file pointer to the beginning of the file and truncates the file to zero length. If file is not found, it creates a new file.
a	Opens file in **write-only** mode. It places the file pointer to the end of the file. If file is not found, it creates a new file.
a+	Opens file in **read-write** mode. It places the file pointer to the end of the file. If file is not found, it creates a new file.
x	Creates and opens file in **write-only** mode. It places the file pointer at the beginning of the file. If file is found, fopen() function returns FALSE.
x+	It is same as x but it creates and opens file in **read-write** mode.
c	Opens file in **write-only** mode. If the file does not exist, it is created. If it exists, it is neither truncated (as opposed to 'w'), nor the call to this function fails (as is the case with 'x'). The file pointer is positioned on the beginning of the file
c+	It is same as c but it opens file in **read-write** mode.

PHP – File Uploading

- PHP file upload features allows to upload binary and text files.
- One can have the full control over the file to be uploaded through PHP authentication and file operation functions.

$_FILES['filename']['name']	Returns file name
$_FILES['filename']['type']	Returns MIME type of the file
$_FILES['filename']['size']	Returns size of the file (in bytes)
$_FILES['filename']['tmp_name']	Returns temporary file name of the file which was stored on the server.
$_FILES['filename']['error']	Returns error code associated with this file

- **move_uploaded_file():** Moves the uploaded file to a new location. The move_uploaded_file() function checks internally if the file is uploaded thorough the POST request. It moves the file if it is uploaded through the POST request.

PHP – Email Basics, Attachments

- PHP has mail()function which is useful in sending the mail from the script.
- **Syntax:**

mail (to, subject, message, headers, parameters)

Where,

to → Represents the address of receiver

subject → Specifies the subject of mail

message → Defines the message which is to be sent

header → This is optional and specifies the additional headers like Cc, Bcc

parameters → This is optional and specifies the additional parameters

Databases with PHP

```
<? Php
$servername = "localhost";
$username = "username";
$password = "password";

//code….

?>
```

Databases with PHP

```php
// Create connection
$conn = mysqli_connect($servername, $username, $password);

// Check connection
if (!$conn) {
  die("Connection failed: " . mysqli_connect_error());
}
echo "Connected successfully";


mysqli_close($conn);
```

Databases with PHP

```php
// Create database


$sql = "CREATE DATABASE myDB";
if (mysqli_query($conn, $sql)) {
  echo "Database created successfully";
} else {
  echo "Error creating database: " . mysqli_error($conn);
}
```

Databases with PHP

```
// sql to create table
$sql = "CREATE TABLE MyGuests (
id INT(6) UNSIGNED AUTO_INCREMENT PRIMARY KEY,
firstname VARCHAR(30) NOT NULL,
lastname VARCHAR(30) NOT NULL,
email VARCHAR(50),
reg_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP
)";

if (mysqli_query($conn, $sql)) {
  echo "Table MyGuests created successfully";
} else {
  echo "Error creating table: " . mysqli_error($conn);
}
```

Databases with PHP

```
// Insert Query
$sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com')";


if (mysqli_query($conn, $sql)) {
  echo "New record created successfully";
} else {
  echo "Error: " . $sql . "<br>" . mysqli_error($conn);
}
```

Databases with PHP

```
// Select Query
$sql = "SELECT id, firstname, lastname FROM MyGuests";
$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {
  // output data of each row
  while($row = mysqli_fetch_assoc($result)) {
    echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
$row["lastname"]. "<br>";
  }
} else {
  echo "0 results";
}
```

Databases with PHP

```
// Select Query
$sql = "SELECT id, firstname, lastname FROM MyGuests WHERE lastname='Doe'";
$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {
  // output data of each row
  while($row = mysqli_fetch_assoc($result)) {
    echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
$row["lastname"]. "<br>";
  }
} else {
  echo "0 results";
}
```

Databases with PHP

```
// Select Query
$sql = "SELECT id, firstname, lastname FROM MyGuests ORDER BY lastname";
$result = mysqli_query($conn, $sql);


if (mysqli_num_rows($result) > 0) {
  // output data of each row
  while($row = mysqli_fetch_assoc($result)) {
    echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
$row["lastname"]. "<br>";
  }
} else {
  echo "0 results";
}
```

Databases with PHP

```php
// sql to delete a record
$sql = "DELETE FROM MyGuests WHERE id=3";

if (mysqli_query($conn, $sql)) {
  echo "Record deleted successfully";
} else {
  echo "Error deleting record: " . mysqli_error($conn);
}
```

Databases with PHP

```php
// sql to update a record
$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";

if (mysqli_query($conn, $sql)) {
  echo "Record updated successfully";
} else {
  echo "Error updating record: " . mysqli_error($conn);
}
```